

Universality and first deviation for orthoscheme reflection groups in every dimension

Vico Bonfioli

Abstract

The right-triangle universality of the companion paper — all shapes share a growth prefix, then deviate at a shortest-vector law — is the two-dimensional case of a general dichotomy. We isolate the abstract mechanism: for any family of quotients of a *common* group, spherical growth is shape-independent through a horizon equal to half the shortest added relator, and first deviates exactly there. We then exhibit the orthoscheme (path-simplex) reflection groups as a family realizing this dichotomy in *every* dimension: generic shapes are rigid (all give one universal growth sequence), and every shape agrees with it up to a finite, shape-dependent horizon. The universal sequences and their ratios $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$ are computed for $n = 2, 3, 4$; the arithmetic of the universal sequence is dimension-dependent — transcendental at $n = 2$ (OEIS A396406), rational at $n = 3$ ($9 \cdot 2^{d-2}$) — so *universality is general while transcendence is special to the plane*.

1 The abstract dichotomy

Let $G = \langle S \rangle$ be a group with a finite symmetric generating set S , and let ℓ_G denote word length. For a normal subgroup $N \trianglelefteq G$ the quotient $Q = G/N$ inherits the generating set $\bar{S}$; write $s_d(H) = \#\{h \in H : \ell_H(h) = d\}$ for the sphere sizes and $K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$ for the length of the shortest nontrivial element of N .

Theorem 1 (Universality–deviation dichotomy). *Let $n := \lceil K(N)/2 \rceil$. Then*

$$s_d(Q) = s_d(G) \quad \text{for all } d < n, \quad \text{and} \quad s_n(Q) < s_n(G).$$

Equivalently, the first depth at which the growth of the quotient falls below that of G is exactly $\lceil K(N)/2 \rceil$.

Proof. The projection $\pi : G \rightarrow Q$ is surjective and 1-Lipschitz, so $\ell_Q(\pi g) \leq \ell_G(g)$ and π maps the radius- d ball $B_d(G)$ onto $B_d(Q)$. Thus $s_d(Q) \leq s_d(G)$ for every d , with equality iff π is injective on the sphere $\Sigma_d(G)$ and no two elements of $\Sigma_d(G)$, $\Sigma_{d'}(G)$ with $d' \leq d$ collapse together — i.e. iff $\pi|_{B_d(G)}$ is injective and length-preserving.

Suppose $d < n$, i.e. $2d < K(N)$. If $\pi g = \pi h$ with $g, h \in B_d(G)$, then $gh^{-1} \in N$ and $\ell_G(gh^{-1}) \leq \ell_G(g) + \ell_G(h) \leq 2d < K(N)$, forcing $gh^{-1} = 1$, so $g = h$. Hence $\pi|_{B_d(G)}$ is injective; it is also length-preserving, since $\ell_Q(\pi g) < \ell_G(g)$ would give a shorter representative g' with $\pi g' = \pi g$, whence $g(g')^{-1} \in N \setminus \{1\}$ has length $\leq 2d < K(N)$, again impossible. Therefore $s_d(Q) = s_d(G)$ for all $d < n$, and cumulatively $|B_{n-1}(Q)| = |B_{n-1}(G)|$.

At depth n : choose $g_0 \in N$ with $\ell_G(g_0) = K(N)$ and split a geodesic word for it as $g_0 = w_1 w_2$ with $\ell_G(w_1) = \lceil K/2 \rceil = n$ and $\ell_G(w_2) = \lfloor K/2 \rfloor$. Both subwords are geodesic (a shorter one would shorten g_0), and $\pi w_1 = \pi(w_2^{-1})$ because $w_1 w_2 \in N$; yet $w_1 \neq w_2^{-1}$ in G (else $g_0 = 1$). Thus two distinct elements of $B_n(G)$ share a fiber, so $|B_n(Q)| < |B_n(G)|$; combined with $|B_{n-1}(Q)| = |B_{n-1}(G)|$ this gives $s_n(Q) = |B_n(Q)| - |B_{n-1}(Q)| < s_n(G)$. $\square$

Remark 2. The theorem is a statement about *one* quotient. It becomes a *universality* statement when applied to a *family* $\{Q_T = G/N_T\}$ sharing the common group G : every member has the same growth as G — hence as every other member — through its own horizon $n_T = \lceil K(N_T)/2 \rceil$, and members with larger $K(N_T)$ agree longer. The growth is shape-independent exactly as far as the shortest added relator allows, and the first deviation depth is a shortest-vector problem in N_T . This is the abstract content of the right-triangle theorems of the companion paper, where G is the generic (virtually lamplighter) group, $N_T = 2(t-1)\mu_T\mathbb{Z}[t^{\pm 1}]$, and n_T is the deviation-depth law.

2 The orthoscheme family in every dimension

An *n-orthoscheme* (path simplex) is a Euclidean simplex with vertices $P_0, \dots, P_n$ whose consecutive edges $P_{k-1}P_k$ are mutually orthogonal; normalizing $P_k = P_{k-1} + \ell_k e_k$ places edge k along the axis e_k with leg length $\ell_k > 0$ ($\ell_1 = 1$). Its $n+1$ facets carry reflections $R_0, \dots, R_n$ (the reflection R_j in the facet opposite P_j), generating a group $W_T \leq \text{Isom}(\mathbb{R}^n)$; for $n = 2$ this is the right-triangle side-reflection group of the companion paper. We measure the geodesic growth $s_d(W_T)$: the number of distinct isometries of word length d in $\{R_0, \dots, R_n\}$.

Two facets R_i, R_j with $|i-j| \geq 2$ are orthogonal, hence commute; consecutive facets meet at a dihedral angle that, for generic legs, is an irrational multiple of π . As with right triangles, this makes the *generic* orthoscheme group rigid: its isomorphism type, and hence its growth, is independent of the (generic) shape. Computation confirms this and pins the universal sequences.

Proposition 3 (Generic universality, computational). *For $n \in \{2, 3, 4\}$, all tested generic n -orthoschemes share one geodesic-growth sequence $U^{(n)}$, and the successive ratios converge to $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$:*

$$\begin{aligned} n = 2 : \quad U^{(2)} &= 1, 3, 5, 8, 13, 21, 34, 55, 89, 144, 225, \dots & (\text{OEIS A396406}) \\ n = 3 : \quad U^{(3)} &= 1, 4, 9, 18, 36, 72, 144, 288, \dots = \{1, 4\} \text{ then } 9 \cdot 2^{d-2} & (r_3 = 2, \text{ exact to depth 16}) \\ n = 4 : \quad U^{(4)} &= 1, 5, 14, 33, 75, 169, 380, 854, 1919, 4312, 9689, \dots & (r_4 = 1 + 2 \cos \frac{2\pi}{7} = 2.24698 \dots) \end{aligned}$$

The agreement is over four generic shapes per dimension, to depth 14 ($n = 2$), 16 ($n = 3$), 12 ($n = 4$); the ratio $r_3 = 2$ and $r_4 = 2.24698$ are reproduced to full precision. (Script code/zeta_probe/orthoscheme_univ)

Proposition 4 (Finite-horizon deviation, exact). *Special shapes agree with $U^{(n)}$ up to a shape-dependent depth and then deviate. In dimension 3, in exact rational arithmetic:*

$$\begin{aligned} (1, 1, 1) : \quad & 1, 4, 9, \mathbf{17}, \dots \text{ deviates at depth 3; } & (1, 1, 2), (2, 1, 1) : 1, 4, 9, 18, \mathbf{35}, \dots \text{ depth 4; } \\ (1, \sqrt{3}, 1) : & \text{ depth 6; } & (1, 2, 3) : \text{ agrees past depth 14 (deep deviation).} \end{aligned}$$

The deviations are certified in exact arithmetic (rational legs give rational reflections), so they are genuine group-theoretic collapses, not numerical artifacts. As in the plane, shapes with a rational dihedral angle deviate early, while shapes with only “generic” angles (e.g. $(1, 2, 3)$) agree far past the reach of direct enumeration.

3 What is proven, and the significance

Theorem 1 is unconditional and dimension-free: it reduces the entire universality-then-deviation phenomenon to two group-theoretic inputs — a common ambient group G , and the shortest-relator length $K(N_T)$ of each shape’s kernel. For right triangles both inputs are supplied in closed form

by the companion paper (the lamplighter identification and the ideal $2(t-1)\mu_T\mathbb{Z}[t^{\pm 1}]$), making the $n = 2$ instance a theorem outright, deviation law and all. Propositions 3–4 establish the two inputs *computationally* in dimensions 3 and 4: the generic group is rigid (a universal sequence exists), and each shape’s horizon is finite. What remains open beyond $n = 2$ is the structural identification of the generic orthoscheme group G_n and of the kernels N_T — the higher analogue of the lamplighter theorem — which would upgrade Propositions 3 and 4 to theorems and yield a closed-form horizon n_T in every dimension.

The phenomenon is thus *general*: a whole family of reflection groups, in every dimension, grows identically until an arithmetic obstruction of the individual shape switches on. Against this backdrop the plane is special in one precise respect — *arithmetic*. The universal sequence is transcendental only at $n = 2$ (A396406, growth rate the transcendental $\beta_2 = 1.4916\dots$, strictly below $r_2 = \varphi$); at $n = 3$ it is the rational $9 \cdot 2^{d-2}$, and the ratios $r_3 = 2$, $r_4 = 2.24698\dots$ govern cleanly. Universality is the rule in every dimension; transcendence is the exception, confined to the plane.